



CodeGloFix Gym Access System

Final Full Installation, Previous Install Removal, Backup, Restore & Fresh System Guide

Company: CodeGloFix

Developed by: AI Engineer Lakshan L.

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Architecture: PyQt-only live display · local backend · browser admin panel

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1. Purpose of this guide

This document is the single, complete reference for deploying and maintaining the CodeGloFix Gym Access System on a customer PC. It covers:

- New / fresh installation
- Previous install removal (complete)
- Backing up a previous install before removal
- Reinstalling while keeping and reusing the old database
- A completely fresh install with the old database removed
- Restoring previously exported data after a reinstall
- Verification and troubleshooting

Read Section 10 (Decision guide) first if you are unsure which path applies.

2. Final architecture summary

- **PyQt-only live attendance display.** The outside 7-inch screen is driven by the PyQt6 live-display wrapper (camera feed + access result popup).
- **Chrome kiosk live view removed.** There is no Chromium kiosk for the live view — only PyQt.
- **Admin panel opens manually in a browser**, login-gated, never on the public display.
- **Backend runs locally** as a single-worker service.
- **Main backend URL:** `http://127.0.0.1:5000/`
- **Admin URL:** `https://localhost/admin` (production) or `http://127.0.0.1:5000/admin` (local test).
- **All persistent data** is stored under `/var/lib/codeglofix`.

3. Important paths

Item	Path
Source path (dev)	<code>/home/auto/Desktop/display/gym_face_access</code>
Installed app	<code>/opt/codeglofix/app</code>
Installed venv	<code>/opt/codeglofix/venv</code>
Data directory	<code>/var/lib/codeglofix</code>
Database	<code>/var/lib/codeglofix/gym.db</code>
Face enrolments	<code>/var/lib/codeglofix/enrolments/shared</code>
Gate config	<code>/var/lib/codeglofix/gate_config.json</code>
Backup folder	<code>/var/backups/codeglofix</code>
Logs	<code>/var/log/codeglofix</code>
Environment file	<code>/etc/codeglofix-access.env</code>

Services: `codeglofix-access@<user>.service` (backend), `codeglofix-live-display.service` (PyQt display), `codeglofix-backup@<user>.{service,timer}`, `codeglofix-health.{service,timer}`.

4. Before removing a previous install — BACK UP FIRST

This is the recommended first step for any change on an existing PC.

Local backup (database + embeddings + gate config + secrets)

```
cd /opt/codeglofix/app
./backup_db.sh
```

Portable export bundle (for moving data to another PC)

```
cd /opt/codeglofix/app
./deploy/export_codeglofix_data.sh
```

The export produces `codeglofix_data_<host>_<date>.tar.gz` containing:

- `gym.db`
- face embeddings / enrolments (`arcface_embeddings.npz` + any face images)
- `gate_config.json`
- an env `SAMPLE` (no secrets) for reference

■ **WARNING** — Never remove or wipe the old database before a backup unless the customer explicitly confirms they do NOT need the old members, face data, plans, payments, access logs, or gate configuration. Deletion of `/var/lib/codeglofix` is permanent.

5. Remove previous install but KEEP old data

This removes the app, services, desktop icons, and nginx setup, but **preserves** `/var/lib/codeglofix` — including the old `gym.db`.

```
cd /opt/codeglofix/app
./deploy/remove_previous_install.sh --dry-run # preview only
./deploy/remove_previous_install.sh          # perform removal
```

- `--dry-run` prints every action without changing anything.
- The default removal does NOT delete `/var/lib/codeglofix`.
- The old database remains available for the next install.
- Previous data (members, enrolments, logs, plans, payments) is reused automatically by the next install.
- The uninstaller also OFFERS to back up the data dir first; pass `--no-db-backup` only if you already have a backup, and `--yes` to auto-confirm.

6. Reinstall using the old database

This is the important "keep old data" path. If `/var/lib/codeglofix/gym.db` already exists, the installer **preserves it** and the refreshed app reuses it (the app file sync explicitly excludes `gym.db`, `gym.db-wal`, and `gym.db-shm`).

```
cd /home/auto/Desktop/display/gym_face_access
./deploy/install_customer_pc.sh
```

Then verify

```
cd /opt/codeglofix/app
./deploy/verify_pyqt_display.sh
./deploy/verify_pyqt_display_install.sh
curl -s http://127.0.0.1:5000/api/health
```

- Old members remain.
- Old face enrolments remain.
- Old access logs remain.
- Old payment / plan data remains if present.
- App files are refreshed to the new version.
- The database is NOT deleted.

■ **NOTE** — If, after a reinstall on a very old database, the Admin Plans tab shows an error, the preserved membership_plans table may predate a newer column. See Troubleshooting (Section 17) for the read-only schema check. Never delete the customer database to "fix" this.

7. Restore old backed-up data after reinstall

Use the import script to restore a previously exported bundle:

```
cd /opt/codeglofix/app
./deploy/import_codeglofix_data.sh /path/to/codeglofix_data_<host>_<date>.tar.gz
```

The import safely: stops the service, backs up the current /var/lib/codeglofix to /var/backups/codeglofix, restores gym.db + enrolments/shared + gate_config.json from the bundle, fixes ownership/permissions, and restarts the service.

Use this when:

- The database was moved to another PC.
- Old data was exported before the reinstall.
- The customer wants previous members / enrolments restored.

8. Completely remove previous install AND old database (fresh-system path)

■ **DANGER** — The commands below PERMANENTLY DELETE old member data, face enrolments, access logs, plans, payments, and gate configuration — unless you have already taken a backup (Section 4). There is no undo.

Step 1 — remove the install (preview, then run)

```
cd /opt/codeglofix/app
./deploy/remove_previous_install.sh --dry-run
./deploy/remove_previous_install.sh
```

Step 2 — delete the old data and logs

```
sudo rm -rf /var/lib/codeglofix
sudo rm -rf /var/log/codeglofix
```

Optional — only if the customer wants to delete backups too

```
sudo rm -rf /var/backups/codeglofix
```

■ **WARNING** — Do NOT delete /var/backups/codeglofix unless the customer is certain no backup is needed. That folder is your last safety net.

9. Fresh system installation after a full wipe

```
cd /home/auto/Desktop/display/gym_face_access
./deploy/install_customer_pc.sh
```

This creates a new /opt/codeglofix/app, new /opt/codeglofix/venv, new /var/lib/codeglofix, a new gym.db (default plans seeded), a fresh service setup, and a clean system with no old member data.

10. Decision guide — which path to choose

Scenario	Use this path
Customer wants old members/data kept	Backup (§4) → remove install (§5) → reinstall (§6). Do NOT delete /var/lib/codeglofix.
Customer wants old data transferred to another PC	Export (§4) → fresh install (§9/§11) → import bundle (§7).
Customer wants a completely clean system	Backup if needed (§4) → remove install (§5) → delete /var/lib/codeglofix (§8) → install fresh (§9).
Previous install broken but data needed	Backup / export first (§4), then reinstall using old DB (§6).
Previous database corrupted / unwanted	Backup first (§4), then full wipe (§8) and fresh install (§9).

11. Normal install command

```
cd /home/auto/Desktop/display/gym_face_access
./deploy/install_customer_pc.sh
```

Run as the normal desktop user (the script calls sudo itself where needed).

12. Verification commands

```
cd /opt/codeglofix/app
./deploy/verify_pyqt_display.sh
./deploy/verify_pyqt_display_install.sh
curl -s http://127.0.0.1:5000/api/health
systemctl status codeglofix-access@$(whoami).service --no-pager
systemctl --user status codeglofix-live-display.service --no-pager
```

A healthy backend returns JSON from /api/health with ok: true and a small camera_fresh_seconds_ago value.

13. Admin open command

```
cd /opt/codeglofix/app
./kiosk/open_admin.sh
```

- Admin URL (production): <https://localhost/admin>
- Admin URL (local test): <http://127.0.0.1:5000/admin>

The admin panel is login-gated and is never shown on the outside display.

14. PyQt live display — manual command

```
LIVE_DISPLAY_URL=http://127.0.0.1:5000/ \
LIVE_DISPLAY_HEALTH_URL=http://127.0.0.1:5000/api/health \
/opt/codeglofix/venv/bin/python /opt/codeglofix/app/kiosk/live_display.py 0
```

- The trailing 0 selects monitor index 0.
- **Exit:** Ctrl + Alt + Q
- Normally the display starts via codeglofix-live-display.service or the "CodeGloFix Live View" desktop icon; the command above is for manual tests.

15. Service commands

Backend service

```
systemctl status codeglofix-access@$(whoami).service --no-pager
sudo systemctl restart codeglofix-access@$(whoami).service
journalctl -u codeglofix-access@$(whoami).service -e --no-pager
```

PyQt live-display (user) service

```
systemctl --user status codeglofix-live-display.service --no-pager
systemctl --user restart codeglofix-live-display.service
journalctl --user -u codeglofix-live-display.service -e --no-pager
```

16. Performance settings

Current production defaults (set in start.sh, all env-overridable):

Setting	Default	Meaning
STREAM_INTERVAL_SECONDS	0.0556	MJPEG display pacing (~18 fps)
CAMERA_WIDTH	896	Requested capture width
CAMERA_HEIGHT	504	Requested capture height
KIOSK_FG_SCALE	0.88	Foreground scale; leaves a visible blurred border

These affect the **display only** — recognition (model, threshold, detector input 320×320) is unchanged. The camera negotiates the nearest supported mode and falls back to 640×480 only if it rejects the request.

15 fps fallback — if CPU is high on the target PC, revert with env overrides (no code change):

```
STREAM_INTERVAL_SECONDS=0.0667 CAMERA_WIDTH=960 CAMERA_HEIGHT=540
```

■ **WARNING** — Do NOT attempt 20 fps unless a CPU measurement on the target PC confirms safe headroom.

17. Troubleshooting

Plans tab shows Loading / error after a reinstall

The preserved database may have an older membership_plans schema. Check it (read-only — does not modify data):

```
sqlite3 /var/lib/codeglofix/gym.db "PRAGMA table_info(membership_plans);"
```

The current schema has columns: id, plan_name, duration_days, price, description, is_active. If a column is missing, that is the cause. Do NOT delete the database — preserve it and apply the backend fix. The Admin UI now shows a clear message instead of hanging on Loading.

Other issues

- **Backend health failed:** /api/health returns nothing or ok:false. Check journalctl -u codeglofix-access@\$(whoami).service -e.
- **PyQt display does not open:** start the backend first, then run the Section 14 command; confirm PyQt6 is in /opt/codeglofix/venv.
- **Admin login issue:** confirm ADMIN_PASSWORD and SESSION_SECRET in /etc/codeglofix-access.env; local test needs COOKIE_SECURE=false over HTTP.
- **Camera not found:** confirm /dev/video0 exists and VIDEO_SOURCE is correct; check the [CAM] Capture resolution log line.
- **Relay not connected:** check RELAY_PORT; for a bench test with no hardware use RELAY_PORT=/dev/ttyRELAY_NONE (never opens a door).
- **No blur visible:** ensure KIOSK_FG_SCALE=0.88 (or lower); at 1.0 with a widescreen capture the border is nearly invisible.
- **High CPU / low FPS:** use the 15 fps fallback env (Section 16); measure FPS from the /video_feed_kiosk stream, not the camera.
- **Wrong venv / numpy ABI error (numpy.dtype size changed):** InsightFace must run in /opt/codeglofix/venv with pinned numpy==1.26.4; confirm the log shows Using python: /opt/codeglofix/venv/bin/python.

18. Final handover checklist

- ■ Previous data backed up (backup_db.sh / export_codeglofix_data.sh)
- ■ Previous install removed if required
- ■ Old DB preserved OR wiped according to the customer's decision
- ■ Install completed
- ■ Backend health OK (/api/health)
- ■ PyQt display opens fullscreen
- ■ Admin panel opens
- ■ Camera works
- ■ Recognition works
- ■ Relay tested if available

- ■ 7-inch display tested
- ■ Reboot test passed
- ■ Power-cut recovery tested
- ■ PDF guide delivered